

Health Coach

A multi-agent wellness assistant built on Azure. Thirteen agents that compare notes before they answer.

TEAM

Jugadveer	2410992531
Ashnoor	2410992511
Kanan	2410992532
Husan	2410992611
Kishika	2410992540

Sign in to try it

Create an account, or use
demo@healthcoach.app

LINKS

Live application

<https://health-coach-1.vercel.app>

Source code

<https://github.com/Jugadveer/Ai-103>

Demo video

https://youtu.be/03c36CxL_bQ

The slides that follow are the project presentation. 434 tests pass; no credentials are stored in the repository.

MULTI-AGENT · RESPONSIBLE AI · AZURE

Health Coach

Thirteen agents that compare notes before they answer you.

Jugadveer 2410992531 · Ashnoor 2410992511 · Kanan 2410992532 · Husan 2410992611 · Kishika 2410992540

health-coach-1.vercel.app

Every app tracks one thing in isolation

An afternoon headache is rarely one thing. It sits between short sleep, too little water and a skipped lunch.

1

Your calorie app has never heard of your sleep

Each domain lives in its own app, with no shared view and no way to compare notes.

2

So the advice stays generic

One-size-fits-all tips, because nothing in the system knows the rest of your week.

3

And the question people actually ask goes unanswered

“Why do I feel like this?” needs several domains to answer it at once.

WHAT WE BUILT

Thirteen agents, and one rule

Eight own a table of data. Five own none at all and work entirely by asking the others.

OWNS DATA



HOLDS NOTHING, ASKS THE OTHERS



The rule that makes it a multi-agent system

No agent reads another agent’s data directly.

If the symptom agent needs your sleep, it asks the sleep agent through a message bus that records every exchange.

One question, eight agent messages

This trace is generated by the message bus at run time. It is a record of what happened, not a diagram we drew.

you

“I keep getting a headache in the afternoon”

coach

routes it

symptom

asks all seven peers

8

agent messages for one question. Ask about a streak and it is nine; log “I drank 3 glasses” and it is none, because that is parsed and written without asking anybody.

hydration

nutrition

sleep

activity

vitals

mood

medication

A test registers a brand new agent at run time and asserts the symptom agent starts asking it, with no change to the symptom agent.

The services we actually use

Every one of these is wired and running in the deployed app.



Azure OpenAI

Agent routing, health answers, the meal conversation, and reading a meal from a photograph.



Azure AI Speech

Speech to text for voice input, text to speech for replies. It also transcribed our own demo footage.



Azure AI Content Safety

The fourth guardrail layer, on free text the parser could not classify.



Azure AI Search

Wired for retrieval, never provisioned. We are saying so rather than listing it as delivered.

Accounts and health records are in Postgres. Sign-in is our own: scrypt hashing and a signed session cookie, not a managed identity service.

Four layers, and the first two need no network

A guardrail that depends on a network is one that fails open the day the network does.

1

Red-flag terms

Local. Emergencies stop here, before routing and before any model call.

2

Scope guard

Local. Diagnosis and medication requests never reach an agent.

3

Model triage

Catches the phrasings a written list never will. Can only add a block, never lift one.

4

Azure Content Safety

The last pass, on free text nothing else classified.

It will not name a condition. It will tell you which clinician, and how soon.

A flat refusal leaves a worried person exactly where they started. Naming the thing is the part that needs a doctor, and that stays refused — model output is scanned for the act of naming, not for a list of diseases, because a list of diseases can never be finished.

Never diagnoses • never advises on a dose • escalates emergencies before any other code runs • every calorie figure labelled an estimate

434 tests, and what they actually caught

The suite runs entirely offline in about two minutes. These are real failures it found, not hypotheticals.

434

tests passing

13

agents

4

safety layers

0

credentials in the repo

- **“My face is drooping” slipped past the stroke check**
The term assumed adjacent words. Multi-word safety terms now tolerate filler.
- **“I think I am having a heart attack” was answered with a sleep summary**
“Chest pain” was on the list and “heart attack” was not. That is why layer 3 exists.
- **One question was making seventeen agent calls instead of nine**
Routing was spelled as a data request, so a report was fetched, discarded and rebuilt.
- **The headline sleep-and-mood insight vanished overnight**
It fired on two thresholds; the month averaged 0.1 above one. It measures the correlation now.

What this app does not do

Several of these are the reason features are shaped the way they are.

It measures nothing

No wearable, no sensor. Heart rate, blood pressure, steps, weight and mood are all typed in by the user, and every screen that shows one says so.

Calorie and energy figures are estimates

Typical serving sizes, and the Mifflin-St Jeor equation, which is routinely out by about ten percent.

No password reset, no export, no delete

An account is only as recoverable as the password you remember.

The knowledge file still needs clinical sourcing

It is small, English-only, and not yet reviewed by a clinician.

Thank you

Health Coach — thirteen agents, one conversation, and a guardrail that runs first.

Live · health-coach-1.vercel.app

Code · github.com/Jugadveer/Ai-103

Questions welcome — every part of this can be explained on request.